Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Ocean Health Chl MEDSEA OceanColour

Issue: 5.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	15/01/2021	All	Creation of the document. The content has been extracted and	V. Brando, M. Sammartino, S. Colella,	R. Santoleri
2.0	10/09/2021	All	Update for EIS December 2021	S. Colella, M. Sammartino	R. Santoleri
3.0	2022/06/16	All	November EIS 2022 – New Template	Simone Colella	V. E. Brando
4.0	2023/06/16	All	November EIS 2023	Simone Colella	V. E. Brando
5.0	2024/07/16	All	November EIS 2024	Simone Colella	V. E. Brando

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I INTRODUCTION

I.1 Product name

Product name	Dataset name
OMI_HEALTH_CHL_MEDSEA_OceanColour_trend	omi_health_chl_medsea_oceanclour_trend
OMI_HEALTH_CHL_MEDSEA_OceanColour_area_averaged_mean	omi_health_chl_medsea_oceanclour_area_averaged_mean
OMI_HEALTH_CHL_MEDSEA_OceanColour_area_averaged_anomaly	N/A

I.2 Indicator evaluation method

The CMEMS Ocean Colour OMIs are derived from the reprocessed (MY) regional gridded chlorophyll products generated and distributed by the CMEMS OC-TAC. The input dataset are spectral remote sensing reflectances (Rrs) derived from multi-sensor (SeaStar-SeaWiFS, AQUA-MODIS, NOAA20-VIIRS, NPP-VIIRS, Envisat-MERIS and Sentinel3-OLCI) (at 1 km resolution) processor and produced by CNR using an in-house processing chain. To date, data are based on R2022.0 processing for NASA/NOAA sensors (SeaStar-SeaWiFS, AQUA-MODIS, NOAA20-VIIRS, NPP-VIIRS), 3rd reprocessing for Envisat-MERIS and baseline collection 3.01 for Sentinel3-OLCI. Chlorophyll product is obtained by means of the Mediterranean Ocean Colour regional algorithms: an updated version of the MedOC4 (Case 1 (off-shore) waters, Volpe et al., 2019, with new coefficients) and AD4 (Case 2 (coastal) waters, Berthon and Zibordi, 2004). The processing chain and the techniques used for algorithms merging are detailed in the corresponding QUID (Colella et al., 2023).

The MY regional chlorophyll dataset was used to compute anomaly map, with respect to reference climatology (1997-2014), daily and monthly regional mean time series and trend map.

Trend map was obtained by applying Colella et al. (2016) methodology. Monthly chlorophyll maps, derived from MY data, are used to compute the trend for the period 1997-2023. Trend analysis has been performed by using the X-11 seasonal adjustment procedure (Pezzulli et al. 2005). The X-11 procedure assumes that the CHL time series Y_t can be decomposed into its three components: the seasonal signal S_t, the inter-annual component T_t, and an irregular component I_t:

$$Y_t = T_t + S_t + I_t$$

The deseasonalized time series is obtained through an iterative process (Colella et al., 2016), by using successive filters, from simple moving-average, seasonal moving-average and Henderson filters (Vantrepotte and Mélin, 2009). At the end of this process, a seasonal signal is given, which is subtracted to the original times series in order to obtain a deseasonalized signal. Then, to estimate the magnitude of trend and its significance, the Mann-Kendall test (Mann, 1945;

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Kendall, 1975) and Sens's method (Sen, 1968) are subsequently applied on the deseasonalized time series. The final trend map is given in terms of relative changes (i.e., % yr^{-1}) by dividing, for each pixel, the trend slope by the corresponding climatological chlorophyll value.

Product: OMI_HEALTH_CHL_MEDSEA_OceanColour_trend

This product includes the trend map obtained by using the methodology described in Colella et al. (2016). Here, trend magnitude and its significance are estimated by applying the Mann-Kendall test and the Sens's method on the deseasonalized monthly time series, as obtained from the X-11 technique, to estimate, trend magnitude and its significance. The trend is expressed in % $year^{-1}$ that represents the relative changes (i.e., percentage) corresponding to the dimensional trend [mg m^{-3} y^{-1}] with respect to the chlorophyll climatology (1997-2014). Only significant trends ($p < 0.05$) are included.

Product: OMI_HEALTH_CHL_MEDSEA_OceanColour_area_averaged_mean

This product includes the monthly mean time series obtained by computing the average of 2D monthly mean (weighted by pixel area) over the region of interest. The deseasonalized time series is obtained by applying the X-11 seasonal adjustment methodology on the original time series as described in Colella et al. (2016), and then Mann-Kendall test (Mann, 1945; Kendall, 1975) and Sens's method (Sen, 1968) are subsequently applied to obtain the magnitude of trend.

Product: OMI_HEALTH_CHL_MEDSEA_OceanColour_area_averaged_anomaly

The regional annual chlorophyll anomaly is computed by subtracting a reference climatology (1997-2014) from the annual chlorophyll mean, on a pixel-by-pixel basis and in log10 space.

I.3 CMEMS products used as input for the computations

Product name	Dataset Name	Documentation
OCEANCOLOUR_MED_BGC_L4_MY_009_144	cmems_obs-oc_med_bgc-plankton_my_l4-multi-1km_P1M	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEM OC-QUID-009-141to144-151to154.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEM OC-PUM.pdf

II REFERENCES

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Colella, S., Brando, V.E., Cicco, A.D., D'Alimonte, D., Forneris, V., Bracaglia, M., 2023. Quality Information Document. Copernicus Marine Service. OCEAN COLOUR PRODUCTION CENTRE, Ocean Colour Mediterranean and Black Sea Observation Product. (https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OC-QUID-009-141to144-151to154.pdf).

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Sen PK. 1968. Estimates of the regression coefficient based on Kendall's tau. J Am Statist Assoc. 63:1379–1389.

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